

Shaping Online Dialogue: Examining Community Rules and Discussion Structures on Reddit

Anna Fang
Human-Computer Interaction
Institute, School of Computer Science
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
amfang@andrew.cmu.edu

Wenjie Yang
Hong Kong University of Science and
Technology
Hong Kong, China
wjyccs@gmail.com

Haiyi Zhu
Human Computer Interaction
Institute
Carnegie Mellon University
Pittsburgh, Pennsylvania, USA
haiyiz@cs.cmu.edu

Abstract

Community rules play a key part in enabling or constraining the behaviors of members in online communities. However, little is known regarding how and to what degree rule changes actually affect community dynamics. In this paper, we explore Reddit to understand how explicit rules may shape the structure of group discussions. Our work uses the top 2000 communities on Reddit over a 1.5 year period and a unique methodology pipeline including (1) establishing a rule taxonomy of behavior-governing rules across Reddit, (2) network analysis to measure discussion structures, and (3) conduct propensity score matching to observe how rule changes may affect network structure. Our study aims to contribute to proactive forms of moderation, and offers several implications for community designers, such as how greater clustering occurs in communities with greater structural regulation and that restriction on allowable content does not reduce the volume of user activity.

CCS Concepts

• **Human-centered computing** → **Empirical studies in collaborative and social computing.**

Keywords

online communities, social computing, moderation, Reddit, network analysis

ACM Reference Format:

Anna Fang, Wenjie Yang, and Haiyi Zhu. 2025. Shaping Online Dialogue: Examining Community Rules and Discussion Structures on Reddit. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3706599.3720188>

1 Introduction

Online communities have become increasingly integral in people's everyday lives for building social relationships and discussing any topic of interest. However, an ongoing challenge in the creation and maintenance of these communities is moderating unwanted user behaviors, such as preventing online harassment or misinformation [7, 33]. While most large platforms (e.g. Facebook, YouTube, Reddit)

engage in multiple levels of content moderation, perhaps the most fundamental element of online community moderation is community rules. Community rules can encode platform and community values [18, 67], form desired behaviors [18, 35], and act as both a guide and explanation for moderators' decisions [56]. Evidence suggests that community-generated rules are particularly effective for regulating users' behaviors, are easiest for new members to adopt, and are context-sensitive [18, 29, 49, 56]. However, empirical and quantitative evidence is lacking regarding how community rules relate to actual user behavior and community dynamics, including contributions to the community and structural patterns among users [23, 40, 50]. As a result, we explore:

RQ: How do changes in rules relate to changes in communities' discussion and user interaction structures, particularly in terms of posting networks (e.g. volume of posting, distribution of comments) and user interaction structures (e.g. active members, average interactions with other users, user clustering)?

To investigate this, we use **Reddit**, which is the 7th most-visited website in the US and has over 130,000 active online sub-communities called "subreddits". Although Reddit includes platform-wide policies, its primary source of governance is each subreddit's rules created by their own community moderators; these rules are usually explicitly written in the subreddit's sidebar. Reddit has a rich and community-driven rule system that has yielded a wide variety of successful online community research [12, 15, 24, 25, 44] and enables us to account for variability in rules across communities that operate within the same unified platform affordances.

We show that significant relationships exist particularly in regards to increases in rules governing the structural components of posting, including significantly greater clustering in the user networks and a more even distribution of connections among users overall. Our study showing how changes in rules relate to shifts in online dialogue presents both a research contribution to the online moderation space as well as practical implications for community stakeholders; community designers and moderators can gain insight that inform decision-making processes to foster online environments aligned with intended goals.

2 Related Work

We situate our study within the rising interest of HCI research in online governance and moderation. In particular, our study takes a unique perspective in studying online governance using a social network analysis perspective to add to a multifaceted understanding of online community dynamics.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3720188>

2.1 Rules and Moderation of Online Communities

Although online community moderation encompasses many elements, our work tackles arguably one of the most foundational structures underlying user behavior in online communities: community rules. Rules serve as guides for moderators and can protect community shared values [10, 11, 27, 29, 58]. While high-level policies like Terms & Conditions often fail to actually govern user behavior, "community-level" rules crafted by local subcommunities have proven to be more effective for fostering adherence and curbing undesirable behaviors [18, 36, 41, 49]. While community-level rules can be informal (e.g. accepted norms), formalized rules like those on Reddit [12] are effective at helping users understand allowable behavior, open up community participation in rule formation, and encourage adherence to normative behavior [18, 29, 49]. Although there exists literature [29] reviewing the effectiveness of rules from a qualitative perspective, our work provides quantitative, empirical evidence of the relationship between these "community-level" rules and user behavior.

We build upon previous work including by Fiesler et al. that has characterized the types and frequency of subreddit rules at scale, finding both context-dependent and sitewide commonalities across Reddit [18]. We draw from Fiesler et al.'s findings for our methods in this study, given the existing categorization is unfit to be used in isolation for behavior-governing studies such as ours given its categorization is primarily content-based rather than behavior-based and varied granularity (broad such as "Prescriptive" versus "Restrictive" to more specific ("Hate Speech")) thus complicating direct comparison. Due to these reasons, our work's methods consider and build upon Fiesler et al.'s classification by providing separate lower- and high-level Reddit rule classification through a behavior-based lens, which can be utilized for studies such as our own to view the effects of rules on user behaviors and patterns.

2.2 Social Network Analysis and Online Community Dynamics

There is a well-established history of network analysis methods applied to online community research [4, 8, 28, 30, 37, 46, 68]. Although there has been a shift in HCI and CSCW scholarship towards language analysis methods for analyzing online community behavior, analyzing *structure* has been and continues to be one of the most important elements in understanding how communities organize and evolve over time topologically through revealing interaction pattern at both a micro- and macro-level [21, 50]. Social network analysis in the modern age of social media has been used to analyze and tackle prevalent issues that can be structurally addressed [7, 17, 43, 47, 51, 60, 63]. Our study in particular uses overall network metrics, established by prior work [38, 48, 59, 65, 66], as a way to measure and compare user dynamics on Reddit.

Although prior work has looked at various online community dynamics through a social network analysis lens, our work fills a central gap of examining online community *governance* using networks. Network analysis as reviewed above has an established history of examining evolution of social network behavior and community dynamics, yet a notable oversight remains in tackling how governance affects community structure. However, there is

evidence that external stimuli does indeed affect community dynamics and structures in meaningful ways [31, 45, 53]. In this same vein, our work investigates the relationship between a community's network structure and one of the most prevalent external forces of change to community dynamics: rule changes.

3 Methods

We explore our research question using a *network* approach, drawing from previous literature on online community users' behaviors [2, 8, 14, 20, 32], and through the lens of "rule changes" – comparing the network structures of subreddits experiencing rule changes with similar subreddits sans rule changes over the same time period. To do this, our work introduces a unique methodology pipeline (Figure 1). In **Stage 1**, we find rule changes over time through establishing and validating a rule taxonomy that is then used to classify rules of subreddits and their changes over time (Section 3.1). In **Stage 2**, we calculate network metrics of the Reddit communities, related to user activity, discussion size, and equality of discussions (3.2). In **Stage 3**, we employed propensity score matching and regression to observe how rule changes may affect network structure (3.3).

3.1 Rule Changes

To extract rule changes, we utilized the digital internet archive tool Wayback Machine, since the official Reddit API does not log community rules nor historical information about subreddits. Wayback Machine allows users to access historical snapshots of internet pages. For our study, we attempted to collect all available archives of the top 2000 subreddit homepages from the Wayback Machine between April 1, 2018 and August 31, 2019. Since snapshot availability depends on manual user requests, 1033 out of 2000 communities did not have any snapshot data; in total, we gathered 347,254 snapshots over the 1.5-year timeframe from 967 unique subreddits (see Appendix for full statistics). 487 subreddits had any rule changes. After obtaining the snapshots, we used regular expressions to parse rules from these archived pages.

3.1.1 Taxonomy of Rules on Reddit. Although other rule taxonomies for Reddit exist [12, 18], they are not fit as is for our study given focus on norms rather than explicit rules, lack of consistent granularity in categorization, or not categorizing rules from a behavioral standpoint. As a result, we built a taxonomy of behavior-governing rules through thematic analysis [6] of 799 community rules from a random sample of 100 subreddits of our dataset (see Appendix). We continued coding until saturation; we organized 56 codes into 15 axial-codes and finally three main themes: **Community Values**, **Content Restrictions**, and **Structural Regulation**. To validate this classification, two independent coders categorized 100 random rules into one of the three themes with "substantial" agreement (Cohen's kappa = 0.722 [13, 42]). Table 1 shows our final rule categories and examples of specific rules is shown in Appendix.

3.1.2 Classifier for rule themes. Using the above rule taxonomy, we built a three-class classification model that allows us to identify changes in the three rule themes across time. We annotated 701 Reddit rules for the training set (57% content restrictions, 34% community values, and 9% structural regulation). The data was then converted into a 5000-dimensional vector using a bag-of-words

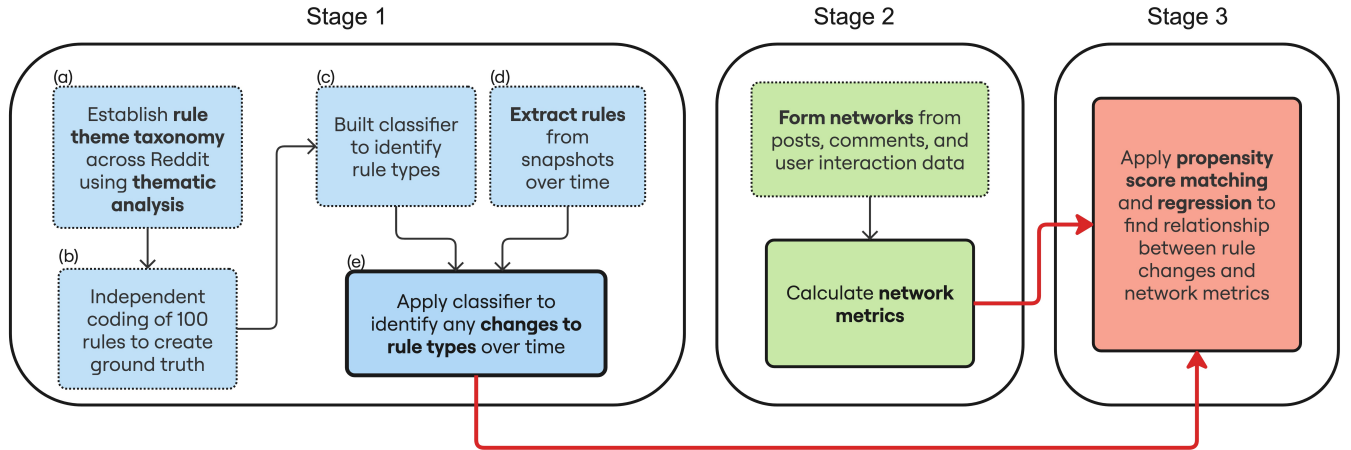


Figure 1: Workflow diagram. We identify rule changes through thematic analysis and classification, calculate network metrics of interest, and PSM and regression to find the relationship between the two.

Table 1: Rule codebook, including the number of subreddits that had a rule in each rule category.

Theme	Rule	# sub-reddits
Community Values	Discourages divisiveness	22
	Enforces respect for others	87
	Avoid distressing material	40
	No promotion of bad behavior	18
	Enable peer feedback	10
Content Restrictions	Posts must be verifiable	23
	No jokes	26
	Content must be original	41
	No promotional content	52
	Allow promotional content, with approval	66
	Posts must be on-topic	79
Structural Regulation	Posts must be high-quality	5
	Requires a minimum text	23
	Requires template for posts	33
	Limits use of title versus description text	2

model. We trained a Logistic Regression classifier (other attempted models’ results available in Appendix) on 80% of the training set, achieving average F1 scores of 0.78, 0.83, and 0.66 for community values, content restrictions, and structural regulation, respectively. Through applying this classifier, we thus discerned rule types and their counts for a given subreddit over time.

3.1.3 Automatic Rule Extraction and Change Detection. For rule change detection, we tracked additions or removals of each rule type between consecutive subreddit snapshots. When a rule difference was detected (i.e., the second snapshot shows a rule change compared to the first), the pre-rule change is the period before the first snapshot while post-rule change is the period after the second. Since archives depend on user requests rather than routine crawls, snapshot intervals can vary, making the precise timing of changes difficult to detect. As a result, we excluded changes with intervals exceeding 60 days. Overall, we identified 1,468 rule changes

across 487 of the 967 subreddits with snapshots, primarily involving content restriction rules, followed by community values and structural regulation-related rules. Detailed statistics are available in the Appendix.

3.2 Network Analysis

Given that the Reddit API does not provide historical post and comment data, we used the expansive Reddit archive dataset called PushShift¹, which is a public data archive that has logged Reddit submissions, comments, and user data in monthly data dumps since 2015 [5].

Using PushShift, we constructed two networks to compare subreddits before and after rule changes:

¹https://github.com/pushshift/google_bigquery

Table 2: Network metrics in our study, our interpretation of their meaning for our study’s context, and prior relevant work outlining the interpretation of the respective metric in a social network analysis setting.

	Dependent Variable & Definition	Network Metric	Prior Work
Submissions Network	Volume of Posts: number of posts (“submissions”)	Vertex count	[48]
	Volume of Comments: number of comments on all posts	Edge count	[48]
	Distribution of comments: measure of inequality for comments distributed among posts	Gini coefficient, in-degree centrality	[59, 65]
User Interaction Network	Active users: number of active users	Vertex count	[38, 48]
	Average interaction: average number of connections that a user had with other users, cohesion of a network	(Average) degree centrality	[19, 64]
	Distribution of user interaction: measure of inequality for relationships distributed among users	Gini coefficient, degree centrality	[59, 65]
	User Clustering: tendency for users to form tightly-knit groups	(Global) clustering coefficient	[16, 66]

- **User Interaction Network.** A directed graph where nodes represent subreddit users, and weighted edges denote interactions (comments on another user’s post or comment). While users can upvote or downvote posts, these actions are anonymous and not analyzable.
- **Submission Network.** An unweighted, directed graph where nodes represent posts (whether original submissions or comments) in the subreddit, and edges are directed from a post to its parent post.

Then, we calculated signals of user interaction and submission network structures using metrics from past literature and of potential interest to moderators/designers as shown in Table 2. We then used a matching-based method to find a control sample for each of the “treated” networks and conducted regression analysis to identify significant relationships between rule changes and network metrics.

3.3 Propensity Score Matching

Similar to prior work done on Reddit [10], we conducted a quasi-experimental analysis using propensity score matching [9, 52] to find causal evidence between rules and network outcomes. We used the 487 subreddits that had any rule change occurrences as a treated group and selected its control as a subreddit that did not have any rule change during the treated subreddit’s observation period and had the most *co-posting* of users – that is, the subreddit that had the most number of active users who posted in both the control and treated subreddit [10] to control for important confounding characteristics. After choosing control subreddits, our team did a manual review of 10 subreddits to verify the treatment and control subreddits were similar on aspects like topic.

After matching, we conducted fixed-effects regression on the treatment-control pairs to assess how rule changes affected network metrics over time. Our panel data represented each change point in community rules (e.g., +1 Structural Regulation, -1 Content Restrictions). Each group is set to be the treatment and respective control subreddit. The regression’s dependent variable is the change in network metrics (pre- vs. post-rule change), while independent variables were changes in rule implementation (e.g., +1 Structural

Regulation, +0 Community Values, -1 Content Restrictions). We also added number of subscribers in each subreddit as a control variable.

4 Results

Given that we ran 21 statistical tests, there is an increased likelihood of spurious significant results. As a result, we used family-wise error rate adjustment through the Bonferroni correction [1] and show results (Table 3 and Table 4) for both the standard 0.05 as well as more stringent Bonferroni threshold of 0.002. We focus our discussion on findings that had statistical significance below the Bonferroni threshold, given their reliability.

Overall, we found that no significant effects on the number of active users (**Model 1**) or number of comments (**Model 3**) were seen with changes in community values, content restriction, or structural regulation rules. Moreover, **Model 4** and **Model 5** show no significant results under the Bonferroni threshold.

There is a significant positive association between **changes in structural regulation rules and the number of posts** ($\beta = 0.179, p < 0.002$) in **Model 2**. This indicates that an increase in structural regulation rules corresponded to a significant rise in the number of posts made within the subreddits. This significant finding makes intuitive sense when considering that most structural rules direct members to find topics of interest easier. Greater organization of a subreddit may be related to greater posting activity as people not only find topics of interest but also develop greater investment to a sub-community of people they more regularly interact with [22]. Additionally, structural regulation rules dictating formatting of posts to a template may lower barrier to entry for new members in the posting process and thus lead to more willingness towards contribution [39].

Greater structural regulation rules were associated with more even distribution of user interaction throughout the subreddits in **Model 6** ($\beta = -0.010, p < 0.002$). For **Model 7**, we found that greater structural regulation rules was significantly related to higher user clustering ($\beta = 0.048, p < 0.002$). For both these results, we hypothesize that findings are due to emergence of clustering

Table 3: Fixed-effect regression results for posting activity models, including active users, number of posts, commenting activity, and distribution of comments per post. Significant results are shown as * $p < 0.05$ and ** $p < 0.002$ (bonferroni method).

	Model 1 # of (active) users	Model 2 # of posts	Model 3 # of comments total	Model 4 Gini coeff., comments per post
	β (SE)	β (SE)	β (SE)	β (SE)
comm. val.	0.028 (0.056)	0.061 (0.044)	0.003 (0.036)	0.003 (0.002)
content restr.	0.025 (0.042)	-0.001 (0.030)	0.015 (0.025)	-0.003* (0.001)
struct. reg.	0.092 (0.087)	0.179** (0.069)	-0.027 (0.056)	-0.004 (0.003)

* $p < 0.05$ ** $p < 0.002$ **Table 4: Fixed-effect regression results for user interaction models, including the average interaction among users, the distribution of this interaction, and the clustering of users. Significant results are shown as * $p < 0.05$ and ** $p < 0.002$ (bonferroni method).**

	Model 5 Average interaction (per user)	Model 6 Distribution of user interaction	Model 7 User clustering
	β (SE)	β (SE)	β (SE)
comm. val.	0.799* (0.401)	-0.001 (0.001)	0.001 (0.001)
content restr.	0.573* (0.291)	0.001 (0.001)	0.000 (0.001)
struct. reg.	0.318 (0.656)	-0.010** (0.002)	0.048** (0.002)

* $p < 0.05$ ** $p < 0.002$

even within "sub"reddits (i.e. a sub-sub-community); structural regulation leads to a restriction of space where people are allowed to post. Enforcing structure may enable users to find content they are interested in and thus form local groups/clusters around that topic. In other words, organized structure makes it easier for users to interact more with a clique of similar users. Moreover, this finding may be of interest to community designers as studies have found that this narrowing of the space leads to a higher level of investment, productivity, and support from members [22]. Additionally, this aligns with prior work that has found smaller communities tend towards group-level dynamics (Model 6) [22]. A qualitative look at the subreddits that increased structural regulation revealed that most of these communities revolved around broad subjects (e.g. r/portland) that may benefit from directing certain types of posts to certain threads or time periods (i.e. dedicated threads to buying/selling items).

5 Discussion

5.1 Takeaways for Community Designers

Structural rules showed the most significance in governing user behavior structures. This may be due to structural rules being significantly more tangible and unambiguous. Rules like requiring minimum text or post templates leave little room for interpretation, making them easier for users to follow and moderators to enforce.

Moreover, violations of these rules are objective and straightforward to govern. Structural rules may also be obeyed more closely because of platform scaffolds themselves rather than just actions of moderators. On Reddit specifically, interface functionalities can be adjusted to align with structural rules; for example, a post can be prevented from being submitted unless the submission system identifies it has passed a minimum text threshold.

Significant findings that may be relevant for moderators is that, while increasing structural regulation may increase posting activity and user interaction, it also may create greater subgroup formation within the community. For many community creators and moderators, this change that may come from adding structural rules may counter their goals, such as leading to more siloed interactions rather than fostering a unified group culture. Conversely, interpreting our study's results at the standard significant level (0.05) rather than the stringent Bonferroni correction (0.002) may support the notion that those aiming to mitigate clustering effects could instead adopt rules on community values or content restrictions given findings of heightened user interaction levels without accompanying increase in clustering (see Model 5 and Model 7). However, we also note that rule-setting and community behavior can be a cyclical process rather than unidirectional; our study provides evidence towards a causal theory on rule-setting affecting community behavior, but community behavior can also influence rule formation [57].

Contrary to expectations, we found that Content Restrictions did not negatively affect user interaction or posting metrics. Although these rules may be seen as the most strict and restrictive, such as disallowing certain content and media and enforcement of information sources, they may not demotivate participation. This follows prior work that has found content moderation may even encourage contribution, because it clarifies a community's goals [34]. One common rule within Content Restrictions is requiring on-topic content, which aligns with Arguello et al.'s findings that focused posts increase reciprocal actions like comments [3]. Additionally, while Sassenberg found that narrow topics promote group identity [54], our study do not find support for any differences in how interactions are distributed in the community given a rise in Content Restrictions.

5.2 Toward Proactive Forms of Moderation

Traditional approaches to moderating online communities have typically centered on a "reactive" approach, addressing or mitigating the harm from undesirable behaviors after they have already impacted a community. For example, popular moderation techniques in online platforms and studied in HCI research have been banning [26], deleting content [62], and quarantining [10, 61] – techniques that take effect on violating behaviors that have already occurred. By contrast, our work studying the effects of behavior-governing rules contributes to a largely unexplored but promising design space of *proactive* and *self-regulatory* moderation. This emerging design space of promoting positive behavior or otherwise intervening to violating content before it spreads in a community has gained traction recently [55, 56]. As Seering notes, "*Among the potential research directions for interventions, this space is perhaps the most wide-open; the design space for tools that proactively shift community behaviors in more positive directions is constrained mostly only by the creativity of interested researchers*" [56]. While community-led moderation, like Reddit's, has shown potential to promote good behavior [56], many existing attempts (e.g., comment pre-screening) are not scalable, especially in large communities with limited moderators [55]. Proactive moderation strategies that scale effectively are critical for larger platforms, and our study demonstrates that explicit community rules may have effectiveness at guiding behavior toward intended dynamics at scale.

5.3 Limitations

Our work has the following limitations. First, while we ensured data saturation through qualitative review, a larger sample may be required to draw broader causal conclusions about rule effects across Reddit's 100,000+ active communities. We also acknowledge classifier accuracy limitations, particularly for structural regulation rules (66% accuracy), likely due to the lower frequency of these rules. However, we note that our classifier accuracy is similar to that of prior work [18]. Lastly, while this study offers a method pipeline that may be useful for other research in accessing historical Reddit data, we also note that our study's use of Wayback Machine with third-party archives still dealt with 51% of subreddits lacking snapshot data, which may bias results as smaller or niche subreddits may be underrepresented. Nonetheless, we observed no significant correlation between missing data and subreddit size,

possibly because our sample was limited to the top 2000 subreddits that are inherently large or well-established communities. Overall, the issue of missing historical data remains a challenge due to Reddit's limited API support.

References

- [1] Hervé Abdi et al. 2007. Bonferroni and Šidák corrections for multiple comparisons. *Encyclopedia of measurement and statistics* 3, 01 (2007), 2007.
- [2] Manju K Ahuja and Kathleen M Carley. 1999. Network Structure in Virtual Organizations. *Organization Science* 10, 6 (Dec. 1999), 741–757.
- [3] Jaime Arguello, Brian S Butler, Elisabeth Joyce, Robert Kraut, Kimberly S Ling, Carolyn Rosé, and Xiaoping Wang. 2006. Talk to me: foundations for successful individual-group interactions in online communities. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Montréal, Québec, Canada) (CHI '06). Association for Computing Machinery, New York, NY, USA, 959–968.
- [4] Eytan Bakshy, Itamar Rosenn, Cameron Marlow, and Lada Adamic. 2012. The role of social networks in information diffusion. In *Proceedings of the 21st international conference on World Wide Web*. 519–528.
- [5] Jason Baumgartner, Savvas Zannettou, Brian Keegan, Megan Squire, and Jeremy Blackburn. 2020. The Pushshift Reddit Dataset. *ICWSM* 14 (May 2020), 830–839.
- [6] V Braun and V Clarke. 2012. Thematic analysis. (2012).
- [7] Ceren Budak, Divyakant Agrawal, and Amr El Abbadi. 2011. Limiting the spread of misinformation in social networks. In *Proceedings of the 20th international conference on World wide web* (Hyderabad, India) (WWW '11). Association for Computing Machinery, New York, NY, USA, 665–674.
- [8] Cody Buntain and Jennifer Golbeck. 2014. Identifying social roles in reddit using network structure. In *Proceedings of the 23rd International Conference on World Wide Web* (Seoul, Korea) (WWW '14 Companion). Association for Computing Machinery, New York, NY, USA, 615–620.
- [9] Marco Caliendo and Sabine Kopeinig. 2008. Some practical guidance for the implementation of propensity score matching. *J. Econ. Surv.* 22, 1 (Feb. 2008), 31–72.
- [10] Eshwar Chandrasekharan, Shagun Jhaver, Amy Bruckman, and Eric Gilbert. 2022. Quarantined! Examining the Effects of a Community-Wide Moderation Intervention on Reddit. *ACM Trans. Comput.-Hum. Interact.* 29, 4 (March 2022), 1–26.
- [11] Eshwar Chandrasekharan, Umashanthi Pavalanathan, Anirudh Srinivasan, Adam Glynn, Jacob Eisenstein, and Eric Gilbert. 2017. You Can't Stay Here: The Efficacy of Reddit's 2015 Ban Examined Through Hate Speech. *Proc. ACM Hum.-Comput. Interact.* 1, CSCW (Dec. 2017), 1–22.
- [12] Eshwar Chandrasekharan, Mattia Samory, Shagun Jhaver, Hunter Charvat, Amy Bruckman, Cliff Lampe, Jacob Eisenstein, and Eric Gilbert. 2018. The Internet's Hidden Rules: An Empirical Study of Reddit Norm Violations at Micro, Meso, and Macro Scales. *Proc. ACM Hum.-Comput. Interact.* 2, CSCW (Nov. 2018), 1–25.
- [13] Jacob Cohen. 1960. A coefficient of agreement for nominal scales. *Educational and psychological measurement* 20, 1 (1960), 37–46.
- [14] Srayan Datta and Eytan Adar. 2019. Extracting Inter-Community Conflicts in Reddit. *ICWSM* 13 (July 2019), 146–157.
- [15] Munmun De Choudhury and Sushovan De. 2014. Mental Health Discourse on reddit: Self-Disclosure, Social Support, and Anonymity. In *Eighth International AAAI Conference on Weblogs and Social Media*.
- [16] Giorgio Fagiolo. 2007. Clustering in complex directed networks. *Physical Review E* 76, 2 (2007), 026107.
- [17] Youssra Fettach and Lamia Benhiba. 2020. Pro-Eating Disorders and Pro-Recovery Communities on Reddit: Text and Network Comparative Analyses. In *Proceedings of the 21st International Conference on Information Integration and Web-based Applications & Services* (Munich, Germany) (iiWAS2019). Association for Computing Machinery, New York, NY, USA, 277–286.
- [18] Casey Fiesler, Jialun Jiang, Joshua McCann, Kyle Frye, and Jed Brubaker. 2018. Reddit Rules! Characterizing an Ecosystem of Governance. *ICWSM* 12, 1 (June 2018).
- [19] Dale Ganley and Cliff Lampe. 2009. The ties that bind: Social network principles in online communities. *Decision support systems* 47, 3 (2009), 266–274.
- [20] Laura Garton, Caroline Haythornthwaite, and Barry Wellman. 1997. Studying Online Social Networks. *J. Comput. Mediat. Commun.* 3, 1 (June 1997), JCMC313.
- [21] Eric Gilbert and Karrie Karahalios. 2009. Predicting tie strength with social media. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Boston, MA, USA) (CHI '09). Association for Computing Machinery, New York, NY, USA, 211–220. <https://doi.org/10.1145/1518701.1518736>
- [22] Sohyeon Hwang and Jeremy D Foote. 2021. Why do People Participate in Small Online Communities? *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2 (Oct. 2021), 1–25.
- [23] Alicia Iriberry and Gony Leroy. 2009. A life-cycle perspective on online community success. *ACM Computing Surveys (CSUR)* 41, 2 (2009), 1–29.

- [24] Matthew R Jamnik and David J Lane. 2017. The use of reddit as an inexpensive source for high-quality data. , 5 pages.
- [25] Shagun Jhaver, Iris Birman, Eric Gilbert, and Amy Bruckman. 2019. Human-Machine Collaboration for Content Regulation: The Case of Reddit Automoderator. *ACM Trans. Comput.-Hum. Interact.* 26, 5 (July 2019), 1–35.
- [26] Shagun Jhaver, Sucheta Ghoshal, Amy Bruckman, and Eric Gilbert. 2018. Online Harassment and Content Moderation: The Case of Blocklists. *ACM Trans. Comput.-Hum. Interact.* 25, 2 (March 2018), 1–33.
- [27] Jialun Aaron Jiang, Charles Kiene, Skyler Middler, Jed R Brubaker, and Casey Fiesler. 2019. Moderation Challenges in Voice-based Online Communities on Discord. *Proc. ACM Hum.-Comput. Interact.* 3, CSCW (Nov. 2019), 1–23.
- [28] David Kempe, Jon Kleinberg, and Éva Tardos. 2003. Maximizing the spread of influence through a social network. In *Proceedings of the ninth ACM SIGKDD international conference on Knowledge discovery and data mining*. 137–146.
- [29] Sara Kiesler, Robert Kraut, Paul Resnick, and Aniket Kittur. 2012. Regulating behavior in online communities. *Building successful online communities: Evidence-based social design* 1 (2012), 4–2.
- [30] Danielle K Kilgo, Joseph J Yoo, Vinicio Sinta, Stephanie Geise, Melissa Suran, and Thomas J Johnson. 2016. Led it on Reddit: An exploratory study examining opinion leadership on Reddit. *First Monday* (Aug. 2016).
- [31] Jooho Kim and Makarand Hastak. 2018. Social network analysis: Characteristics of online social networks after a disaster. *International journal of information management* 38, 1 (2018), 86–96.
- [32] Jooho Kim and Makarand Hastak. 2018. Social network analysis: Characteristics of online social networks after a disaster. *Int. J. Inf. Manage.* 38, 1 (Feb. 2018), 86–96.
- [33] Robin M Kowalski and Elizabeth Whittaker. 2015. Cyberbullying. In *The Wiley Handbook of Psychology, Technology, and Society*. John Wiley & Sons, Ltd, Chichester, UK, 142–157.
- [34] Robert E Kraut and Paul Resnick. 2011. Encouraging contribution to online communities. *Building successful online communities: Evidence-based social design* (2011), 21–76.
- [35] Robert E Kraut and Paul Resnick. 2012. *Building Successful Online Communities: Evidence-Based Social Design*. MIT Press.
- [36] Cliff Lampe, Paul Zube, Jusil Lee, Chul Hyun Park, and Erik Johnston. 2014. Crowdsourcing civility: A natural experiment examining the effects of distributed moderation in online forums. *Gov. Inf. Q.* 31, 2 (April 2014), 317–326.
- [37] Jure Leskovec, Daniel Huttenlocher, and Jon Kleinberg. 2010. Predicting positive and negative links in online social networks. In *Proceedings of the 19th international conference on World wide web*. 641–650.
- [38] Michael Lieberman. 2014. Visualizing big data: Social network analysis. In *Digital research conference*. 1–23.
- [39] Kimberly Ling, Gerard Beenen, Pamela Ludford, Xiaoqing Wang, Klarissa Chang, Xin Li, Dan Cosley, Dan Frankowski, Loren Terveen, Al Mamunur Rashid, Paul Resnick, and Robert Kraut. 2005. Using social psychology to motivate contributions to online communities. *J. Comput. Mediat. Commun.* 10, 4 (July 2005), 00–00.
- [40] Sanna Malinen. 2015. Understanding user participation in online communities: A systematic literature review of empirical studies. *Computers in human behavior* 46 (2015), 228–238.
- [41] J Nathan Matias. 2016. Posting rules in online discussions prevents problems & increases participation. *Civil Servant* 8 (2016).
- [42] Nora McDonald, Sarita Schoenebeck, and Andrea Forte. 2019. Reliability and inter-rater reliability in qualitative research: Norms and guidelines for CSCW and HCI practice. *Proceedings of the ACM on human-computer interaction* 3, CSCW (2019), 1–23.
- [43] Liz McQuillan, Erin McAweeney, Alicia Bargar, and Alex Ruch. 2020. Cultural convergence: Insights into the behavior of misinformation networks on twitter. *arXiv preprint arXiv:2007.03443* (2020).
- [44] Alexey N Medvedev, Renaud Lambiotte, and Jean-Charles Delvenne. 2019. The Anatomy of Reddit: An Overview of Academic Research. In *Dynamics On and Of Complex Networks III*. Springer International Publishing, 183–204.
- [45] Seth A Myers, Chenguang Zhu, and Jure Leskovec. 2012. Information diffusion and external influence in networks. In *Proceedings of the 18th ACM SIGKDD international conference on Knowledge discovery and data mining*. 33–41.
- [46] M E J Newman. 2004. Detecting community structure in networks. , 321–330 pages.
- [47] Nam P Nguyen, Guanhua Yan, My T Thai, and Stephan Eidenbenz. 2012. Containment of misinformation spread in online social networks. In *Proceedings of the 4th Annual ACM Web Science Conference*. 213–222.
- [48] Marcia Oliveira and Joao Gama. 2012. An overview of social network analysis. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery* 2, 2 (2012), 99–115.
- [49] Elinor Ostrom. 2000. Collective Action and the Evolution of Social Norms. *J. Econ. Perspect.* 14, 3 (Sept. 2000), 137–158.
- [50] Pietro Panzarasa, Tore Opsahl, and Kathleen M Carley. 2009. Patterns and dynamics of users' behavior and interaction: Network analysis of an online community. *Journal of the American Society for Information Science and Technology* 60, 5 (2009), 911–932.
- [51] Irene V Pasquetto, Eaman Jahani, Shubham Atreja, and Matthew Baum. 2022. Social debunking of misinformation on WhatsApp: the case for strong and in-group ties. *Proceedings of the ACM on human-computer interaction* 6, CSCW1 (2022), 1–35.
- [52] Neil Pearce. 2016. Analysis of matched case-control studies. *BMJ* 352 (Feb. 2016), i969.
- [53] Daniel M Romero, Brian Uzzi, and Jon Kleinberg. 2016. Social networks under stress. In *Proceedings of the 25th International Conference on World Wide Web*. 9–20.
- [54] Kai Sassenberg. 2002. Common bond and common identity groups on the Internet: Attachment and normative behavior in on-topic and off-topic chats. *Group Dyn.* 6, 1 (March 2002), 27–37.
- [55] Charlotte Schluger, Jonathan P Chang, Cristian Danescu-Niculescu-Mizil, and Karen Levy. 2022. Proactive Moderation of Online Discussions: Existing Practices and the Potential for Algorithmic Support. *Proc. ACM Hum.-Comput. Interact.* 6, CSCW2 (Nov. 2022), 1–27.
- [56] Joseph Seering. 2020. Reconsidering community self-moderation: the role of research in supporting community-based models for online content moderation. *Proc. ACM Hum.-Comput. Interact* 3 (2020).
- [57] Joseph Seering, Juan Pablo Flores, Saiph Savage, and Jessica Hammer. 2018. The Social Roles of Bots: Evaluating Impact of Bots on Discussions in Online Communities. *Proc. ACM Hum.-Comput. Interact.* 2, CSCW (Nov. 2018), 1–29.
- [58] Joseph Seering, Tony Wang, Jina Yoon, and Geoff Kaufman. 2019. Moderator engagement and community development in the age of algorithms. *New Media & Society* 21, 7 (2019), 1417–1443.
- [59] AK Sen. 1973. On economic inequality Clarendon Press.
- [60] Chengcheng Shao, Pik-Mai Hui, Lei Wang, Xinwen Jiang, Alessandro Flammini, Filippo Menczer, and Giovanni Luca Ciampaglia. 2018. Anatomy of an online misinformation network. *Plos one* 13, 4 (2018), e0196087.
- [61] Qinlan Shen and Carolyn Rose. 2019. The discourse of online content moderation: Investigating polarized user responses to changes in reddit's quarantine policy. In *Proceedings of the third workshop on abusive language online*. 58–69.
- [62] Kumar Bhargav Srinivasan, Cristian Danescu-Niculescu-Mizil, Lillian Lee, and Chenhao Tan. 2019. Content removal as a moderation strategy: Compliance and other outcomes in the changemyview community. *Proceedings of the ACM on Human-Computer Interaction* 3, CSCW (2019), 1–21.
- [63] K Bret Staudt Willet and Jeffrey P Carpenter. 2020. Teachers on Reddit? Exploring contributions and interactions in four teaching-related subreddits. *Journal of Research on Technology in Education* 52, 2 (April 2020), 216–233.
- [64] Sergio L Toral, M Rocio Martínez-Torres, Federico Barrero, and Francisco Cortes. 2009. An empirical study of the driving forces behind online communities. *Internet Research* 19, 4 (2009), 378–392.
- [65] Trevor van Mierlo, Douglas Hyatt, and Andrew T Ching. 2016. Employing the Gini coefficient to measure participation inequality in treatment-focused Digital Health Social Networks. *Network Modeling Analysis in Health Informatics and Bioinformatics* 5 (2016), 1–10.
- [66] Duncan J Watts and Steven H Strogatz. 1998. Collective dynamics of 'small-world' networks. *nature* 393, 6684 (1998), 440–442.
- [67] Galen Weld, Amy X Zhang, and Tim Althoff. 2021. Making Online Communities 'Better': A Taxonomy of Community Values on Reddit. (Sept. 2021). arXiv:2109.05152 [cs.HC]
- [68] Jun Zhang, Mark S Ackerman, and Lada Adamic. 2007. Expertise networks in online communities: structure and algorithms. In *Proceedings of the 16th international conference on World Wide Web* (Banff, Alberta, Canada) (WWW '07). Association for Computing Machinery, New York, NY, USA, 221–230.

Table 5: Descriptive statistics of our study dataset

Number of subreddits	967
Number of subreddits with any rule changes over Apr. 2018 - Aug. 2019	487
Total number of snapshots	347,254
Avg (mean) number of snapshots per subreddit	359
Avg (median) number of snapshots per subreddit	59

6 Appendix

6.1 Full Dataset Statistics

6.2 Subreddits used in thematic analysis for rule codebook

AccidentalRenaissance	compsci	holdmycosmo	notliketheothergirls	succulents
ADHD	Cooking	Homebrewing	obs	sushi
Advice	creepyPMs	HongKong	oilpen	TalesFromRetail
AmateurRoomPorn	CryptoCurrencies	IdiotsInCars	Overwatch	technology
AmericanHorrorStory	csharp	Instantregret	pcmasterrace	terriblefacebookmemes
ar15	Daytrading	ipad	Persona5	thalassophobia
argentina	depression	KamikazeByWords	Pets	thebachelor
AskAnthropology	disney	keto	phoenix	TheRightCantMeme
asktransgender	dogs	legendofkorra	Portland	theydidthemath
awfuleverything	electricians	listentothis	programming	TIHI
aww	ethereum	LofiHipHop	quotes	tipofmyjoystick
battlefield_4	fantasybball	lucifer	rarepuppers	trashy
beauty	fermentation	MagicArena	RobinHoodPennyStocks	virtualreality
blunderyears	FragileWhiteRedditor	magicTCG	runescape	visualnovels
Bonsai	FrugalFemaleFashion	MakeupAddiction	RWBY	warcraftlore
BreakUps	ghibli	minneapolis	Seaofthieves	Warthunder
canada	gifs	motorcycles	ShitMomGroupsSay	westworld
Cartalk	goodyearwelt	MouseReview	singapore	WhatsWrongWithYourDog
CBD	hmm	namenerds	SkincareAddicts	WWE
Coffee	holdmybeer	Nike	StockMarket	youtubehaiku

6.3 Example rules

Theme	Rule	Example	# subreddits
Community Values	Discourages divisiveness	<i>No excessive arguing about social or political topics (r/trashy)</i> <i>No controversial posts (r/KamikazeByWords)</i>	22
	Enforces respect for others	<i>No hate speech (r/gifs)</i> <i>Be nice (r/tipofmyjoystick)</i>	87
	Avoid distressing material	<i>No upsetting content (r/Portland)</i> <i>No videos involving death (r/youtubehaiku)</i>	40
	No promotion of bad behavior	<i>Comments promoting piracy with any links to pirate sites are banned (r/visualnovels)</i>	18
	Enable peer feedback	<i>Report content via modmail or Reddit's reporting system (r/WWE)</i>	10
Content Restrictions	Posts must be verifiable	<i>No unsubstantiated rumors (r/disney)</i> <i>No pseudoscience or misinformation (r/ADHD)</i>	23
	No jokes	<i>No memes (r/sushi)</i> <i>No jokes, memes, or cartoons (r/thalassophobia)</i>	26
	Content must be original	<i>Duplicate or similar post will be removed (r/canada)</i>	41
	No promotional content	<i>No referral links/begging/faucets/casinos/pumps (r/CryptoCurrencies)</i> <i>No begging, trading, or selling (r/pcmasterrace)</i>	52
	Allow promotional content, with approval	<i>Keep promotion reasonable (r/seaofthieves)</i> <i>Follow guidelines about self-promotion (r/Homebrewing)</i>	66
	Posts must be on-topic	<i>Must be relevant to Overwatch (r/Overwatch)</i>	79
	Posts must be high-quality	<i>New accounts with no karma will be removed to weed out spam (r/beauty)</i> <i>Posts or questions lacking substance may be removed (r/coffee)</i>	5
Structural Regulation	Requires a minimum text	<i>Posts must be at least 300 characters (r/ADHD)</i>	23
	Requires template for posts	<i>Playlists can only be posted on weekends (r/LofiHipHop)</i> <i>Follow post template (r/MagicArena)</i>	33
	Limits use of title versus description text	<i>No description use (r/ar15)</i>	2

Table 6: Rule codebook organized by theme, including examples for each axial-code and the number of classified subreddits in the thematic analysis.

6.4 Model accuracy for rule classifier

Model	Accuracy	F1
Logistic Regression	0.76	0.77
Random Forest	0.65	0.71
Decision Tree	0.67	0.67
SVM	0.68	0.72
XGBoost	0.73	0.76

Table 7: Accuracy and F1-score of models for our rule classifier. Our study used the highest accuracy Logistic Regression classifier for analysis.

6.5 Rule change occurrences and types, statistics

Table 8: Descriptive statistics about changes to rule types in dataset, including the number of change occurrences for each type and the mean volume of change (e.g. +2 indicates that a change event in the designated category experiences an average of adding two rules)

	Number (%)	Mean (std)
Change Event	1468	
+ community values	323	1.41 (0.90)
- community values	153	-1.50 (1.04)
+ content restriction	527	1.65 (1.20)
- content restriction	180	-1.44 (1.00)
+ structural regulation	197	1.24 (0.56)
- structural regulation	88	-1.17 (0.53)

6.6 Effects of changing the number of rules

We found that there was a significant increase in clustering with more rules, as well as significantly greater user interaction with more than half an interaction increase with a single increase in the number of rules. Additionally, we found that power distribution between users becomes more uneven. We did not find any significant effects for the submission network.

	Model 1 vertex count	Model 2 edge count	Model 3 density	Model 4 clustering	Model 5 degree, avg	Model 6 degree, gini.	Model 7 betw. centr. gini
	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)	β (SE)
Number of rules	1821 (1071)	16846 (16160)	0.000 (0.000)	0.0053* (0.001)	0.6229* (0.189)	-.00076 (0.001)	.0004439* (0.000)

* $p < 0.05$

	Model 8 vertex count (log) β (SE)	Model 9 edge count (log) β (SE)	Model 10 degree, avg. β (SE)	Model 11 degree, gini. β (SE)
Number of rules	0.039 (0.020)	0.016 (0.016)	-0.001 (0.001)	-0.000 (0.000)

* $p < 0.05$

Table 9: Results for fixed-effect regression for user interaction network (top) and submission network (bottom) using independent variable of number of rules.